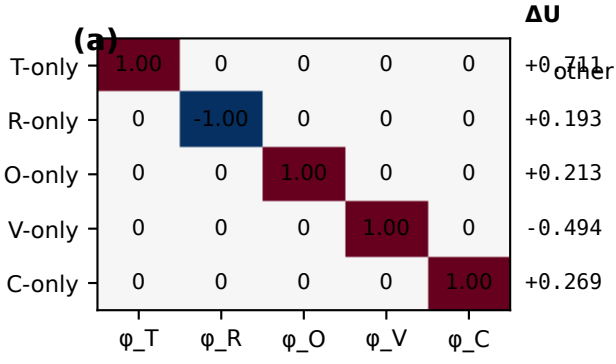

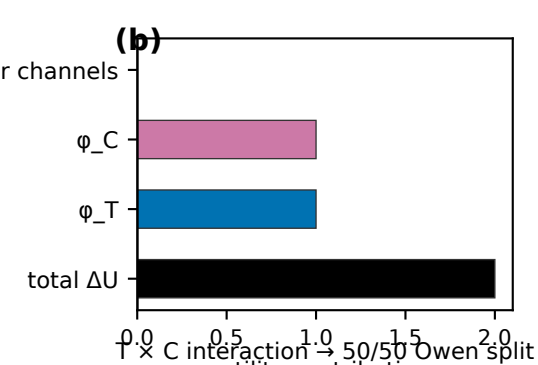

(c) **Validation group** | **max error / status**

Validation group	max error / status
Score → pattern reconstruction	0
Pattern/value → z reconstruction	0
Mask + batch partition	PASS
Independent endpoints	0
Candidate isolation / live leakage	0
Coarse–fine quotient bridge	1.11e-16

Overall validation: PASS